

UNIVERSITY OF TLEMCEN
FACULTY OF SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

Machine Learning
M1 / ING 4
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TP 1: Data Exploration, Manipulation and Visualization

Objective

The objective of this practical work is to introduce the basic steps of data exploration, manipulation, and visualization using Python and the Iris dataset. Students will learn how to load a CSV file, inspect a dataset, select and filter observations, sort and group data, and create visualizations using `pandas`, `numpy`, `matplotlib`, and `seaborn`.

Instructions: Complete the tasks in a Jupyter Notebook or Google Colab. Include your code, outputs, and brief interpretations. Give each plot a title, labeled axes, measurement units where applicable, and a legend when needed.

Iris Dataset

The Iris dataset contains measurements of three Iris species: Setosa, Versicolor, and Virginica. It includes four numerical variables: sepal length, sepal width, petal length, and petal width, as well as a categorical variable representing the species.



Figure 1: The three Iris species: Setosa, Versicolor, and Virginica.

A. Environment Setup and Dataset Introduction

1. Import the following Python libraries: `pandas`, `numpy`, `matplotlib.pyplot`, and `seaborn`. Briefly identify the role of each library.
2. Explain what a CSV file is, then load `iris.csv` into a Pandas DataFrame named `iris_df` using `read_csv()`.
3. Identify what each observation represents. Specify the input features and the target variable. The dataset contains the columns `sepal_length`, `sepal_width`, `petal_length`, `petal_width`, and `species`. Measurements are expressed in centimetres.

B. Data Exploration

4. Display the first five observations using `head()` and the last five observations using `tail()`.
5. Modify the previous commands to display the first ten and the last ten observations.
6. Determine the number of rows and columns using `shape`. Explain the difference between the number of columns and the number of input features.
7. Use `info()` to inspect the column names, data types, and number of non-missing values.
8. Compute descriptive statistics using `describe()`. Identify the minimum, maximum, mean, and standard deviation of `sepal_length`.
9. Count the missing values in each column using `isna().sum()`. Is missing-value treatment necessary for this dataset? Justify your answer using the result.

C. Data Manipulation

10. Select the `petal_length` column and display its first five values.
11. Select the `petal_length` and `petal_width` columns together and display the first five rows.
12. Extract all flowers belonging to the Setosa species and determine their number.
13. Extract all flowers whose petal length is greater than 5 cm.
14. Extract all Virginica flowers whose sepal width is greater than 3 cm. Combine the conditions using the `&` operator.
15. Extract all flowers belonging to either Setosa or Virginica. Combine the conditions using the `|` operator.
16. Sort the dataset by petal length in descending order and display the five flowers with the longest petals.
17. Sort the dataset by petal length in ascending order and display the five flowers with the shortest petals.
18. Calculate the mean petal length for each species using `groupby()`. Round the results to two decimal places and identify the species with the largest mean.

D. Data Visualization

For each visualization, provide a short interpretation (one or two sentences) based on the plot.

19. **Class distribution:** Count the observations belonging to each species and represent the counts using a bar chart. Are the classes balanced? Justify your answer.
20. **Histogram:** Plot the distribution of sepal length using ten bins. Explain what the horizontal axis, vertical axis, and height of each bar represent. Identify the most populated interval.
21. **Scatter plot:** Plot petal length against petal width. Use different colors and marker shapes to distinguish the species. Identify the species that appears clearly separated and the species that overlap.
22. **3D visualization:** Create a 3D scatter plot using sepal length, petal length, and petal width as the three axes. Distinguish the species using colors and a legend. Compare the observed separation with the 2D scatter plot and discuss how the viewing angle may affect interpretation.

Homework

Dataset Description

The Titanic dataset contains information about passengers, including survival status, passenger class, sex, age, family relationships, ticket fare, cabin, and port of embarkation.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

Figure 2: Example of the Titanic dataset displayed as a Pandas DataFrame.

1. Determine the number of observations and variables.
2. Compute the main descriptive statistics for numerical variables.
3. Display the data type of each variable.
4. Determine the number of distinct values for each variable.
5. Determine the number of missing values in each variable.
6. Remove duplicate observations if necessary.
7. Identify the variables containing missing values.
8. Apply an appropriate strategy to handle missing values.
9. Remove the columns `PassengerId`, `Name`, `Ticket`, and `Cabin`.
10. Convert the `Sex` variable into a categorical representation using `to_categorical()`.
11. Apply One-Hot Encoding to the `Embarked` variable using `get_dummies()`.